The Geometry of Action Readiness: A Control-Theoretic Framework for Subjective Experience and Affective Geometry

Abstract

Subjective affective experience presents a persistent theoretical challenge for cognitive science, neuroscience and philosophy of mind: explaining not only why mental states carry positive or negative valence, but why they exhibit distinct, directional motivational geometries - such as the urgent, stressful quality of acute pain, the pull of hunger, the tension of unfulfilled desire, or the effortlessness of relief. Traditional functionalist and identity theories often treat qualia as arbitrary sensory labels appended to neural signals, leaving an explanatory gap between physical computation and qualitative phenomenological character.

Here, we propose a unified, control-theoretic framework that is consistent with Antonio Damasio's theory of homeostatic somatic feelings, Nico Frijda's concept of action readiness, Michael Graziano's Attention Schema / Self-Model theory, and Emily Falk's subjective value valuation. We hypothesize that subjective experience arises when an organism's higher-order self-model constructs a low-dimensional, coarse-grained representation of its own ongoing regulatory and evaluative control dynamics.

Within this framework, we propose that:

1. **Valence** is governed by the temporal derivative of prediction errors on subjective value ($\dot{\epsilon}_{\mathrm{SV}}$), where outcomes better than predicted lead to positive affect and outcomes worse than predicted to negative affect. In addition, valence depends on the current level of subjective value, so that sustained, fully predicted pain or hunger remains unpleasant

2. **Qualitative character and motivational shape** reflect macroscopic fixed points of coarse-grained homeostatic control loops, measured via the **participation ratio (**D_{eff}**)** of the action-readiness distribution, the **attentional capture** κ of attention on its target, and the **orientation** Ω of action readiness relative to that target; and

3. **Qualia** represent the macroscopic properties generated by a coarse-graining process that integrates high-dimensional, micro-scale stochastic neural signals into the self-model's lossy description. It is reframed as the informational geometry of self-correcting control loops

We discuss why the theory helps to understand that physical pain and social pain feel similar. Finally, we discuss how reframing qualia as the informational geometry of self-correcting control loops offers a concrete perspective on the structure of subjective experience.

1. Introduction: The Dual Problem of Affective Experience

A central question in cognitive science and philosophy of mind is why conscious states feel the way they do. While significant progress has been made in mapping the neurobiological correlates of perception and motor control, the affective dimension of subjective experience - the intrinsic "goodness" or "badness" of feelings, along with their specific qualitative textures - remains notoriously difficult to ground in physical mechanisms.

For AIs and philosophers feeling the own existence might seem mysterious, for human babies feeling the own existence is the only thing they do. If you have seen babies crying and screaming, then you cannot come to a different conclusion. Other animals can walk or swim on the first day, humans cannot. They need to vocalize their primordial emotions to survive. For humans feeling the own existence is normal, it accompanies us every day and every single moment. Human babies are pure, raw emotions. In this sense the very first secret of the universe that a baby discovers is feeling the own existence. As we grow older, we start to forget this fundamental insight and try to construct complicated theories instead.

When examining affective qualia, classical theories frequently encounter two problems:

1. **The Valence Problem:** Why are certain neural states intrinsically accompanied by positive or negative subjective value, rather than remaining phenomenologically neutral information readouts? A neural information readout would be perceiving a specific wavelength of light. Physically colors like red, green and blue are just labels to distinguish between different parts of the color spectrum. Each color is associated with a certain, precise wavelength. Red has 625 to 750 nm, green 495 to 570 nm and blue has 450 to 495. The wavelength can be measured physically and objectively. There is nothing subjective about a wavelength.

2. **The Geometric Problem:** Why do affective states possess distinct, directional *motivational shapes*? Pain does not merely signal tissue damage; it exerts an urgent pressure to *stop* the painful event and narrows at the same time the possible actions because any action increases the pain. Hunger does not merely signal a nutrient deficit; it acts as an internal vacuum or external *pull*. Unfulfilled desire manifests as a directional *tension*, while satisfaction is experienced as a reduction of pressure and a sensation of *effortlessness*. Affects and emotions differ in the geometry of the action space they impose. They can be distinguished by narrowing versus broadening, pushing versus pulling, stopping vs continuing

Existing approaches routinely fall short of addressing both dimensions simultaneously. Purely functionalist or computational accounts (e.g., Global Workspace Theory) excel at describing information broadcasting and cognitive access but often reduce qualia to arbitrary functional labels tagged onto sensory processing. Conversely, purely phenomenological or identity theories assert the reality of qualia but struggle to explain why specific neural dynamics generate precise qualitative textures.

We would like to follow the approach of Marc D. Lewis who proposed dynamic systems modeling as a language to close the gap between abstract psychological theories of emotion and concrete, physical brain mechanics (Lewis, 2005). He writes emotions are an amalgam. They are in fact a combination of appraisal (is something good or bad) and action tendencies (what to do about it). They judge if something is good or bad for subject and recommend an action tendencies based on it. In this sense emotions are already a bridge between both problems.

As Mark Solms writes in “The Hidden Spring" (Solms, 2021) the foundational form of consciousness is raw emotion and affect rather than sensory perception. According to Solms affects, emotions and feelings are the only things that are intrinsically conscious. Solms identifies the “hidden spring” of this consciousness as the upper brainstem and subcortical regions. He argues sentient subjectivity is fundamentally constituted by affect. You are conscious because you feel things like pain, fear, panic, or pleasure. Emotions are biological demands that signal when our physiological survival needs are deviating from a safe baseline.

Jaak Panksepp agrees that the foundational form of consciousness is phenomenal and affective. He writes in his book “Affective neuroscience” that the brain was an emotional organ long before it became a thinking one and that raw feelings grow “upward” from ancient subcortical structures (Panksepp, 1998). Jaak distinguishes between three levels of emotions: affect, emotion and feeling, where affect is most animalic, biological part, and feeling the most cultural and social part:

- **Affect**: The most basic, automatic, and pre-conscious dimension of experience
- **Emotion**: The automatic action readiness response to a specific trigger or internal/external event
- **Feeling**: The subjective conscious interpretation, mental experience and personal labeling of an emotion or affect

This article focuses on the biological, animalic part of subjective experience, which is stronger in affects and emotions and weaker in feelings where the social and cultural influence is bigger.

1.1 The Synthesis

To bridge the existing gaps between valence and action tendency, 1st person view and 3rd person view, subjective and objective experience - this includes the explanatory gap from Joseph Levine which describes our inability to understand subjective, conscious experiences from objective, physical descriptions of the brain (Levine, 2001) - we propose that qualitative experience is neither an arbitrary byproduct of neural firing nor an inexplicable non-physical essence. Instead, qualia are the informational topologies of an organism's own regulatory and evaluative control loops as rendered by its higher-order self-model. The self-model is not an observer; its description of the control state and the own homeostatic control loops is what the organism reports as experience.

This framework synthesizes five foundational, previously disparate pillars of cognitive neuroscience and affective psychology:

1. **Homeostatic Regulation (Damasio, 1999, 2010):** The biological foundation. Organisms maintain life through continuous interoceptive monitoring of visceral variables relative to survival set-points. Primary feelings (pain, hunger, temperature) represent the cognitive translation of ongoing somatic regulation.

2. **Evaluative Appraisal & Subjective Value (Falk, 2025):** The economic evaluation. The brain evaluates internal and external states by computing a real-time Subjective Value $\mathrm{SV} = R - E$, balancing expected reward (R) against risk and effort (E)

3. **Action Readiness (Frijda, 1986, 2007):** The vector mechanics. Evaluative states are not static scalars; they manifest as *states of action readiness* - directed vector fields in behavior space that prepare the organism to establish, maintain, or disrupt relationships with its environment.

4. **Self-Modeling & Attention Schemas (Graziano, 2013, 2019):** The rendering engine. The brain constructs a simplified, lossy, low-dimensional internal model of its own processing. When this self-model represents the organism's evaluative and regulatory vectors as attributes of the *self*, subjective experience is born. The self-model represents not only the action-readiness vector but also where attention is directed, and the vector's orientation relative to the attended target.

5. **Perceiving One's Own Emotions (Strout et al, 2004)**: The missing ingredient. Perceiving one's own emotions means your mind perceives the ongoing loop between your environment and your body's readiness to act, generating the conscious "feeling".

2. The Self-Model's Attention-Gated Compression

To transition from qualitative description to a predictive science of subjective experience, we formalize the interaction between homeostatic regulation, evaluative action readiness, and self-model projection. Statistical field theory is a theoretical framework in physics that describes systems with a large or infinite number of interacting degrees of freedom by replacing discrete microscopic variables with continuous fields

2.1 Layer 1: Stochastic Neural Dynamics and Microscopic Noise

The microscopic neural substrate is represented as a high-dimensional, stochastic state vector $\mathbf{M}^{(0)}(x,t) \in \mathbb{R}^N$ governed by coupled Langevin-type Stochastic Differential Equations (SDEs) subject to thermal and physiological noise:

$$\frac{d\mathbf{M}^{(0)}}{dt} = -\nabla_{\mathbf{M}} U\left(\mathbf{M}^{(0)}\right) + \boldsymbol{\eta}(t)$$

where $U\left(\mathbf{M}^{(0)}\right)$ is the energy landscape of neural firing and $\boldsymbol{\eta}(t)$ represents white noise driven by molecular-scale Brownian dynamics.

2.2 Layer 2: Subjective Value and Action Readiness Vector Fields

The organism evaluates prospective actions by computing a real-time Subjective Value (SV) function in a cost-benefit or ROI (return on investment) valuation

$$\mathrm{SV}(t) = R(t) - E(t) = \ w_1\, V_{intrinsic} + w_2\, V_{self} + w_3\, V_{social} - \ w_4\, C_{effort}$$

- $R(t) = w_1\, V_{intrinsic} + w_2\, V_{self} + w_3\, V_{social}$:
  Expected homeostatic reward and intrinsic value $V_{intrinsic}$
  plus self-relevance V_{self} plus social relevance V_{social}
- $E(t) = w_4\, C_{effort}$
  Metabolic costs and computational effort required for corrective action, including risk of structural damage or homeostatic failure.
- w_1, w_2, w_3, w_4 are contextual weights which change dynamically based on your physical and emotional state

Action readiness vectors $\mathbf{A}(t) \in \mathbb{R}^m$ are derived from the coupled error and subjective value manifold:

$$\mathbf{A}(t) = -\nabla_{\mathbf{x}} V\left(\mathbf{e}(t), \mathrm{SV}(t)\right)$$

2.3 Coarse-Graining Operator

The brain cannot represent every unit of its own activity. The self-model therefore builds a lossy description by pooling units in functional space, weighted by attentional gain $\pi(y)$:

$$P_\pi[A](x;\ \lambda)\ =\ \frac{\int W(x-y;\ \lambda)\ \pi(y)\ A(y)dy}{\int W(x-y;\ \lambda)\ \pi(y)\ dy}$$

where λ sets the pooling scale (how much detail is averaged away) and W is a smoothing kernel. Applied recursively across the hierarchy, $M_{\{l+1\}} = P_\pi[M_l]$, this gives a family of descriptions at coarser and coarser scales. Attended components survive into the self-model, while unattended components are integrated out but still shift the background (arousal, mood, gist). Focal π gives fear, and flat π gives anxiety.

Because π depends on the system's own state ($\pi = \mathrm{softmax}(\theta \cdot u_y)$), the map is self-consistent, $M^* = P_\pi[M^*]$. The feedback between urgency and gain produces capture attractors (Lewis, 2005, §3.2). We define urgency u_y here as the unsigned, precision-weighted size of the subjective-value signal for each unit, with a level part and a rate part. For each unit or loop y

$$u_y(t) = \Pi_y(t) \cdot \frac{|SV_y(t)| + \eta \cdot |\varepsilon_{SV}, y(t)|}{s_0}$$

where:

- $|SV_y|$ is the **level term**: how far the loop is from its set-point (the homeostatic error).
- $|\varepsilon_{SV}, y|$ is the **rate term**: how much better or worse than predicted things are going.
- η weights the two terms, and s_0 is a baseline scale, so that θ in $\mathrm{softmax}(\theta \cdot u)$ is dimensionless.
- Π_y is the **precision** (gain) with which the signal enters the self-model.

The variables of the affective geometry (the level term, κ, Ω, the directional costs, Deff) are read out from this description. We do not assume scale invariance. Whether the coarse-graining shows RG-type fixed points and scaling is an open empirical hypothesis (Section 6.2).

Because attention is driven by the system's own state, the pooling map is self-consistent: urgency raises attentional gain, and gain raises urgency. Such positive-feedback loops are the standard route by which self-organizing systems settle into attractors (Lewis, 2005, §3.2). We call these capture state attractors. They persist after the stimulus is withdrawn and resist distraction, and they are left through relief, regained control or a stronger competing urgency.

2.4 Affective Geometry: Metrics and Measures

To quantify the changes in action readiness we define four measures and metrics:

Capture κ which measures how concentrated attention is on the target (body parts, objects, people, thoughts). It is measurable by gaze dwell, reaction-time costs to distractors, or neural decoding of attentional focus.

$$\kappa = 1 - \frac{H(a_t)}{\log n}$$

Orientation Ω is the signed projection of the mean readiness vector onto the target direction which ranges from +1 (approach) to −1 (avoidance).

$$\Omega = \frac{\langle \bar{A}_t, \hat{\tau}_t \rangle}{\|\bar{A}_t\|}$$

Directional cost is the cost (from SV = R − E) of acting toward (E_{toward}) and away from (E_{away}) the attended target. Ω alone cannot distinguish "no preferred direction" from "all directions blocked": persisting pain is the case in which both costs are high and Ω ≈ 0 reflects inhibition, not indifference.

Participation Ratio D_{eff} quantifies the effective dimensionality and the focus of an organism's action-readiness distribution where σ_i are the eigenvalues of the covariance matrix **C o**f the action-readiness vector A:

$$D_{\text{eff}} = \frac{(\sum_i \sigma_i)^2}{\sum_i \sigma_i^2} = \frac{(\text{tr}\,\mathbf{C})^2}{\text{tr}(\mathbf{C}^2)}$$

Its describes the motivational narrowing or expansion of actions and choice variability and measures how many independent directions the organism's readiness varies along. Measured action-readiness variables could be multi-muscle EMG and choice entropy. The mean readiness *direction* is captured separately by Ω, and the distribution of *attention* by κ. D_{eff} is reported with total variance and is comparable only within a fixed measurement setup (same channels, same number of samples). It can be used to separate escape behavior (D_{eff} low, variance high) from physical freezing (D_{eff} low, variance low).

- **Persisting Pain or Grief (Dimensional Collapse):** minimal $D_{\text{eff}} \to 1$, where behavioral degrees of freedom collapse into an inescapable low-dimensional attractor sink or 1D subspace. The STOP signal of pain acts as a constraint on the action space.

- **Flow (Structured Focus):** D_{eff} is tuned and intermediate, balancing concentration with flexible action channels.

- **Relief or Pleasure (Phase-Space Expansion):** maximal $D_{\text{eff}} \gg 1$, where corrective pressure drops and an exploratory galaxy of behavioral opportunities opens up.

κ, Ω, D_{eff}, and the directional costs E_{toward} and E_{away} are hypothesized to be the *relevant parameters* that survive coarse-graining and locate the organism within the structure.

- κ describes how attention is distributed
- Ω indicates the mean readiness direction (the vector towards the target)
- D_{eff} concerns the variability around that mean

3. Trajectories of Valence

3.1 Subjective Value Prediction Errors and Valence Dynamics

Affective valence is governed by temporal derivatives of prediction errors on the subjective value $\epsilon_{\text{SV}} = \text{SV}_{\text{actual}} - \text{SV}_{\text{predicted}}$ which can be written as

$$\dot{\epsilon}_{\text{SV}}(t) = \frac{d}{dt}\Big(\text{SV}_{\text{actual}}(t) - \text{SV}_{\text{predicted}}(t)\Big)$$

The rate term alone cannot be the whole story: a constant, fully predicted pain or hunger would come out neutral, yet both remain unpleasant. We therefore propose that valence has two components: a level term, given by the current subjective value SV (or the homeostatic error), which accounts for the ongoing badness of unresolved deficits and injuries, and a rate term, given by ε_{SV}, which accounts for relief, disappointment and the sharpness of change (compare Carver & Scheier, 1990; Joffily & Coricelli, 2013). Note that ε_{SV} is a signed prediction error (better or worse than predicted); it is not the same as a reduction in the magnitude of prediction error, which concerns the accuracy of the world model.

- **Positive Acceleration ($\dot{\epsilon}_{\text{SV}} > 0$):** When goal attainment or resource acquisition occurs faster or cheaper than predicted, the system experiences accelerated progress, rendered as elation, relief, flow, or joy. More error reduction means better understanding, more clarity and better predictive alignment

- **Negative Acceleration ($\dot{\epsilon}_{\text{SV}} < 0$):** When friction, effort, or risk unexpectedly outstrips projections, the system experiences deceleration, rendered as frustration, anxiety, or despair. Less error reduction and more friction und uncertainty mean less understanding and more prediction violations

3.2 Testable Predictions

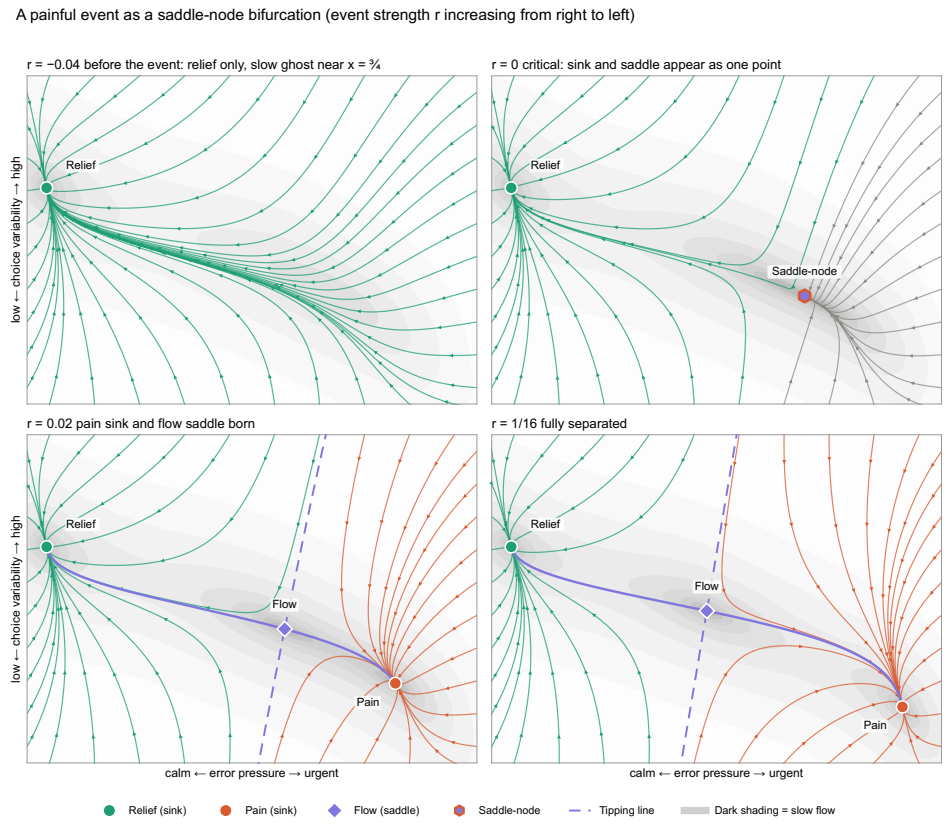
Prediction 1 (orientation and escape). States matched on κ and D_{eff} but differing in the sign of Ω should differ in phenomenology (pulled, pushed, trapped). Such a test needs controls for arousal, and ideally a manipulation of escape feasibility, e.g. giving participants a real or blocked way to terminate an aversive stimulus. If unpleasantness tracks the loss of escape rather than intensity alone, that is a strong, non-obvious result. (see the literature on controllability, Maier & Seligman, 1976).

Prediction 2 (shared control structure of physical and social pain). Pain narrows attention and action, and pleasure broadens them (compare Fredrickson's broaden-and-build, 2004).

Hypothesis: physical and social pain feel similar because they share a control structure: a pressing target (the injured body part or the lost person) that captures attention (high κ) and blocks action in every available direction (high directional cost, $D_{\text{eff}} \rightarrow 1$). They should differ in orientation: grief includes a pull toward an unreachable target ($\Omega > 0$, searching and calling), whereas persisting physical pain has no preferred direction ($\Omega \approx 0$, inhibition).

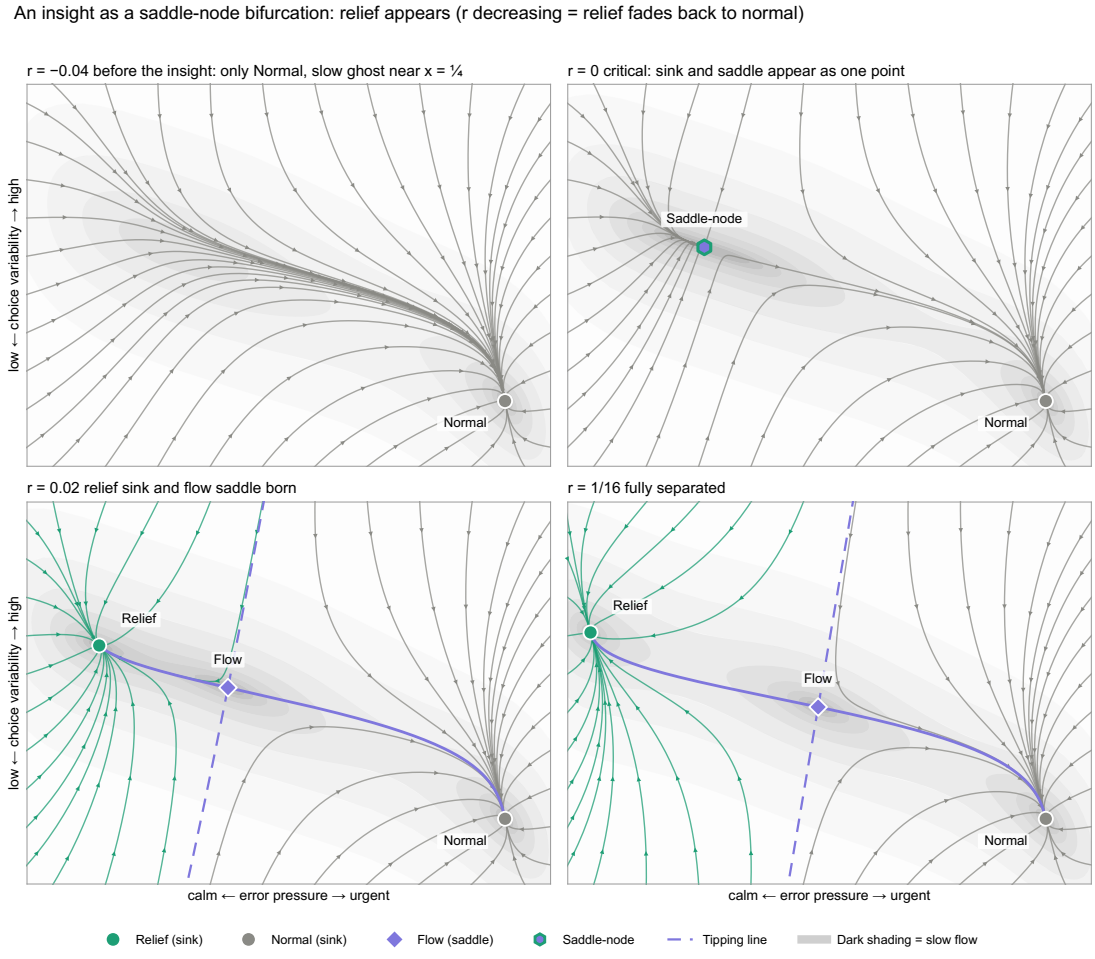
This predicts (a) Behavioral and physiological measures of κ, directional cost and D_{eff} should be similar across the two states and different from a relief state. (b) Ω should differ between them. (c) A decoder trained on one pain type should generalize to the other at coarse spatial scales (for example dACC and insula), and neither should generalize to relief, while at fine multivoxel scales the two types remain separable. Woo et al. (2014) found overlapping dACC/insula activation but separable multivariate patterns, which is consistent with this but does not establish it. Shared coarse activation also has a simpler explanation in general salience or arousal.

If you lose a loved person, your thoughts circle around that person, but there is (almost) nothing you can do to bring the loved person back. It is like a deep sink or little black hole that pulls you in. For physical pain your thoughts circle around the injured part of the body, but there is nothing you can do because every movement of the injured part only increases the pain. In both cases we have low choice variability D_{eff} (because there is nothing we can do) and high urgency / attention capture κ (our thoughts circle around the loss).



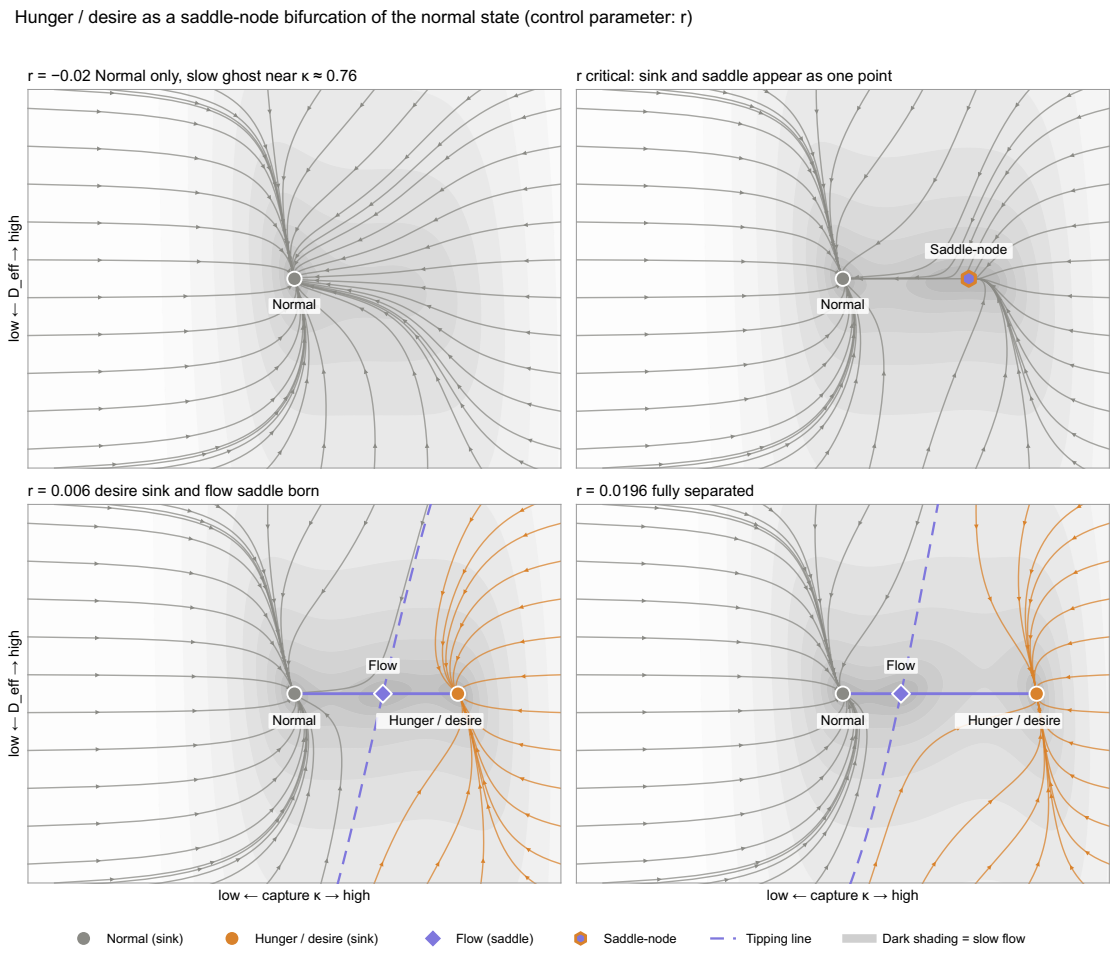
If physical and social pain share a coarse-grained control structure, their neural representations should be separable at fine scales and increasingly indistinguishable as coarse-graining increases, while painful versus relieving states remain distinguishable. Fine-scale differences are treated as analogous to irrelevant operators. Woo et al. (2014) found overlapping dACC/insula activation but separable multivariate patterns, which is consistent with this but doesn't establish it. Shared coarse activation also has a simpler explanation (general salience/arousal).

Prediction 3 (attention, capture and history dependence) Holding nociceptive input constant, manipulating attention (distraction versus focus) should change the geometry in four ways. (a) With the stimulus held constant, increased capture should lower D_{eff} when action is costly in all directions (for example, pain where every movement increases it), but not when action is free (for example, an itch). The prediction is an interaction between κ and directional cost, and not an effect of attention alone. (b) Reported unpleasantness should track κ and the directional cost more closely than input intensity. (c) Capture should show history dependence: once established, it should persist after the stimulus is withdrawn and resist a distractor of a given strength, whereas distraction applied at onset, before capture is established, should be more effective. A purely feed-forward account of attention predicts no such history dependence. (d) The overlap of physical and social pain should depend on κ, being stronger when both capture attention and weaker when they are ignored.



A stressful day in daily life interrupted by joyful insights would look like this saddle-node bifurcation where a fix point for relief appears in the upper left corner where we have high choice variability D_{eff} because there many opportunities and low urgency / error pressure / attentional capture κ.

Hunger & desire can be represented by a fix point for a medium value of D_{eff} (choice variability) and high value of urgency / attentional capture κ.



3.3 Summary of Affective Geometry

By uniting Damasio's interoceptive regulation, Frijda's action readiness, Emily Falk's subjective cost-benefit value valuation, and Graziano's self-modeling framework, we propose an account of why affective states possess distinct motivational structures. Qualia reflect the lossy description of active control loops navigating homeostatic constraints. The most basic emotions

- pleasure and pain (do it – stop it)
- hunger and disgust (consume it – expel it)
- love and fear (approach it – get away from it)

are direct diametrical opposites of primary subjective dimensions.

- fear **pushes** away from threats and activates the organism to prepare it for escaping
- hunger and thirst **pull** towards resources in a search for food and water
- sexual desire **pulls** towards mates because it is a search for reproduction
- relief **eases** because it reduces pressure and broadens focus
- pain is a stressful sink of attention because it increases pressure and narrows focus by **stopping** painful actions

If feelings are the shape of the organism's control strategy as seen by its self-model, the shapes look like this:

- **Fear activates as an escape vector:** The organism is mobilized for rapid physical displacement from a threat. In control space, this manifests as a high-magnitude directional vector pointing away from a danger source, accompanied by hyper-arousal and immediate motor prioritization.

- **Hunger and thirst pull as search attractors:** The system registers a physiological deficit (SV $\ll 0$ via resource depletion E or homeostatic error $\mathbf{e}$). The self-model renders this deficit not as a static number, but as a directed gradient vector (an external or internal pull) drawing exploratory action toward missing resources.

- **Relief eases by broadening focus and reducing pressure:** Following sudden error reduction ($\dot{\epsilon}_{\mathrm{SV}} > 0$), corrective pressure drops precipitously. The Participation Ratio (D_{eff}) expands, unlocking behavioral degrees of freedom and resulting in the phenomenological experience of warmth, ease, and expanded cognitive scope.

- **Pain is stressful because it narrows focus and increases pressure:** Faced with a large homeostatic error or threat to tissue integrity, the system deploys an emergency containment strategy. The Participation Ratio collapses ($D_{\mathrm{eff}} \to 1$), locking behavioral degrees of freedom into an intense, narrow focus that overrides all other motivational drives.

Feeling	Capture κ	Dimensionality D_{eff}	Orientation Ω	Control strategy	Predicted signature
Pain	high	very low	~ 0	Escape/protect (acute); containment (persisting)	D_{eff} drops; acute: brief withdrawal (high variance); persisting: inhibition (low variance, high cost in all directions)
Fear vs anxiety	high	low	< 0	Imminent vs diffuse threat	Fear: fixed gaze; anxiety: wide scanning
Hunger/ Desire	high	low-mid	> 0	Directed search	Mid D_{eff}, falling as food is located
Pleasure/ Relief	falling	rising	n/a	Pressure released	D_{eff} rises, variance falls

In fear you try to prepare to increase the distance from the target, but for persisting pain you cannot do this because the target is your own body, and action in every direction is costly and painful. Pain is an urgent STOP signal that narrows the variability of choices and dimensionality of available actions D_{eff}. What makes it unpleasant is that it creates a strong sink for the neural flow of information: all attention is drawn to the painful body part, but any movement of the body part is prohibited, because it would increase the pain. In effect this is a kind of mental model of the physical injury and the violated physical integrity of the body.

In this sense persisting pain is fear without an escape route, because the threatened object is the self. Grief belongs to the same class: the target is a lost person, and no available action reaches them. Acute withdrawal reflexes are the fear-like phase of pain; the sink appears when withdrawal is no longer possible. Two components should be distinguished: a constraint on the feasible action set (high cost in every direction, captured by D_{eff} and the directional costs) and an attentional attractor (captured by κ). They can dissociate: an itch captures attention with little action restriction, and a well-adapted cast restricts action with little capture. A further prediction is therefore that unpleasantness tracks their conjunction.

For hunger, and desire it is similar. Hunger, thirst and desire in general narrow the action space by narrowing the attention and locking the orientation solely on the objects of desire. Narrowing tracks motivational intensity rather than valence alone: approach-motivated positive states also narrow attention (Gable & Harmon-Jones, 2008), whereas low-approach positive states broaden it. Valence is carried by the level and rate terms of Section 3.1, while κ, Ω, D_{eff} and the directional costs describe the geometry of the motivational state.

Panksepp identified 7 primary emotional systems - SEEKING, LUST, CARE, PLAY/JOY, FEAR/ANXIETY, RAGE/ANGER, PANIC/GRIEF (Panksepp, 1998), which fit to our model.

For a system that is controlled by special control structres (like the seven primary, subcortical emotional-action systems that Pankseep has identified) asking "what it is like" is asking about what it is like to have such an architecture, which means in our framework asking about the geometry of the self-model's control state.

- **SEEKING / LUST:** The dopamine-driven motor of anticipation, curiosity, and enthusiasm corresponds directly to our **search attractors** and positive valence dynamics. It is the drive toward resource acquisition, curiosity, and exploratory flow where the participation ratio (D_{eff}) is tuned and active, and prediction errors are positively accelerated ($\epsilon_{SV} > 0$).

- **FEAR / ANXIETY / RAGE:** The circuit triggered by frustration, containment, or threat corresponds to the **directional vector fields of escape and confrontation**. Fear maps onto a high-magnitude vector pointing away from a threat ($\Omega < 0$, high capture κ), mobilizing the organism for rapid physical displacement. Rage maps onto an aggressive approach vector ($\Omega > 0$) triggered by containment or frustration.

- **CARE:** The survival loop designed to avoid bodily harm and respond to danger represents primary homeostatic and social valuation loops within V_{social} and $V_{\text{intrinsic}}$ (Section 2.2), driving approach, bonding, and reproductive optimization through specific evolutionary reward architectures.

- **PANIC / GRIEF:** The distress network governing social loss, loneliness, and separation distress maps precisely onto our discussion of **social pain and dimensional collapse** (Section 3.2). Just as physical pain grief is a pull toward a target that cannot be reached, so $\Omega > 0$. The circuit acts as an inescapable trap where thoughts circle a wound with nowhere to go ($D_{\text{eff}} \to 1$). Panksepp's PANIC/GRIEF circuit governs separation distress, where the target (a lost loved one) cannot be reached, which creates a deep psychological "black hole" or attractor sink.

- **PLAY / JOY:** Joyful, physical social engagement corresponds to states of **phase-space expansion and behavioral flexibility**, where social engagement broadens action channels and maximizes the exploratory degrees of freedom of D_{eff}.

3.4 The Synthesis

To get a step closer to solving the explanatory gap (Levine, 2001) for the affective core of experience, we offer an identity thesis for elementary affect that

(a) removes the conceivability intuition in this domain
(b) derives the features of the feeling from the geometry
(c) explains why it seems irreducible
(d) yields empirical dissociations and a threshold for which systems have it.

By *self-model* we mean a lossy, attention-weighted description of the organism's own control state that is available for the control of behavior and for report (Graziano, 2019). *Awareness* is

used in this functional sense. An active homeostatic control loop has a geometry: how pressing the situation is (the level term of valence), how strongly attention is captured (κ), in which direction action leads (Ω), and what acting in each direction costs (Effort *E*). When the self-model represents the organism's own situation through this geometry, the organism is aware of its situation as urgent, blocked, open or inviting. It is not aware of its loops or of the evaluation as an evaluation. This is a minimal, situated form of awareness, a proto-consciousness of one's situation.

- **Suffering (persisting pain):** negative level term, high κ and high costs/efforts *E* in all directions. The organism cannot exit the state and cannot attend away from it. "Bad" and "powerless" are the words for this structure as seen from inside the system.
- **Relief and consummatory pleasure:** the level term rises or the deficit is being reduced ($\varepsilon_{SV} > 0$), κ and costs/efforts *E* fall. The organism can exit the state and can attend elsewhere. "Ease" and "effortless" are the words for this structure as lived.
- **Pull states (hunger, desire):** high κ, $\Omega > 0$, low cost toward the target and high cost elsewhere.
- **Push states (fear, disgust):** high κ, $\Omega < 0$, low cost in the escape direction. Fear is worse when that direction is blocked.
- **Grief and panic:** a pull toward a target that cannot be reached, so $\Omega > 0$ with high cost in every available direction.

Following Strout, Sokol, Laird and Thompson (Strout et al, 2004) we state that perceiving one's own emotions means your mind perceives the ongoing loop between your environment and your body's readiness to act, and this is the missing ingredient that is generating the conscious "feeling".

Identity Thesis. An organism has an elementary feeling when its self-model represents its own situation through the geometry of an active homeostatic control loop. This representation is transparent: the geometry is not accessible as a structure, only the situation as urgent or normal, blocked or open, constrained or free. The elementary, irreducible feeling is identical to this state. For these states there is no further fact about what it is like beyond the represented geometry, so a system with the same geometry represented in the same way and no feeling is not coherent.

The elementary feeling is a minimal form of core consciousness, a pre-reflective awareness of one's own situation as urgent, normal, blocked or open, constrained or free. It is self-involving, since the situation is given as mine, but not self-reflective, because it requires no self-concept, body concept or narrative self. Systems with control loops but no self-model (a chemotactic bacterium) have the geometry but no feeling. Systems with a self-model but no homeostatic loop (current language models) have a description without anything to describe. Higher emotions add appraisal and reflection on top of this state (Lewis, 2005).

We can describe temperature in various way. Macroscopic as a single number, microscopic as a mean molecular kinetic energy. For subjective attributes like pleasure or pain it is similar. Macroscopic we can describe them as a single number (by asking the subject in question to rate it on a scale from 0 to 10), microscopic as by a lossy compression of microscopic activity.

4. Clinical and Neuropsychological Validation

4.1 Pain Asymbolia: The Decoupling of Evaluation and Self-Model

Pain asymbolia — typically resulting from lesions to the insular cortex, anterior cingulate cortex (ACC), or underlying white matter tracts — presents a striking dissociation where sensory processing remains intact while qualitative affective distress collapses. Patients can accurately locate a pinprick and recognize tissue damage, yet they feel zero urgency, distress, or desire to escape.

- **The Subcortical Processing Split:** Panksepp notes (Panksepp, 1998) that sensory data from bodily damage must activate lower subcortical circuits—specifically the Periaqueductal Gray (PAG) and associated affective networks—to generate true biological "badness" and emotional meaning. Without this subcortical activation, injury is registered merely as neutral, unvalenced information.
- **The Control-Theoretic Interperatation:** Within our framework, pain asymbolia represents a selective disruption of the projection into the self-model between Layer 2 (evaluation and urgency) and Layer 3 (the self-model).
- **The Unrendered Urgency:** While lower interoceptive networks compute high structural risk, the failure to project this emergency vector into the self-model's lossy description means the affective urgency is unrendered. Attention still goes to the injury (κ preserved) so the patient can report the physical sensation, but the directional cost and survival pressure are missing: *"It hurts, but it doesn't bother me."*

4.2 Depersonalization and Derealization Disorders (DPDR)

In depersonalization-derealization disorder (DPDR), patients report a profound, unsettling detachment from their body, mental processes, and external surroundings—often describing their existence as robotic, unreal, or viewed through a film. Within our framework, DPDR represents a global attenuation of the self-modeling gain parameter (γ_M) driven by high-level cortical suppression over deep subcortical affective circuits

- **Subcortical Anchoring (Panksepp's Core SELF):** Panksepp argues (Panksepp, 1998) that an organism's foundational sense of identity and subjective bodily ownership is anchored in a subcortical "core SELF," centered in the midbrain and Periaqueductal Gray (PAG). This primordial circuit generates raw, visceral affective warmth and grounds all conscious states in homeostatic reality.
- **Cortical Insulation & Clamping:** In DPDR—often triggered as a defense mechanism against severe psychological trauma or chronic stress—higher neocortical structures functionally clamp down on these lower subcortical emotional centers.
- **The Control-Theoretic Interperatation:** lower interoceptive regulation and cognitive appraisals continue to function, but the projection into the self-model fails to project the deep subcortical affect. The lossy self-description is deprived of its underlying affective gain ($\gamma_M \to 0$). $\gamma_M \in [0, 1]$ is here the gain with which evaluative signals ($\mathrm{SV}, \varepsilon_{\mathrm{SV}}$) enter the self-model.

Consequently, while the patient can still process sensory data and compute abstract evaluations, their conscious experience is stripped of raw affective coloring. The self-model renders a cold, unanchored description of processing: the system perceives and acts, but because those control loops no longer touch the subcortical core SELF, the experience feels devoid of an "inner light"—leaving an emotional ghost navigating a flat, synthetic world

4.3 Addiction as a Maladaptive Attractor Sink

Clinical disorders of addiction model a pathological distortion of the control manifold, where chronic dopamine-related signals artificially drive high valuation, causing action readiness vectors (**A**) to collapse into a narrow compulsive attractor sink. This means attention is captured by cues through high κ, strongly positive Ω, and narrow D_{eff}.

- **Hijacked Systems**: For Panksepp addiction hijacks the SEEKING system coupled with artificial opioid soothing of the PANIC/GRIEF network (Panksepp, 1998). He explains that drugs of abuse do not merely provide passive "pleasure"; they hyper-activate the dopamine-driven SEEKING circuit. This triggers an intense, chronic state of anticipation and frantic craving, locking all of the brain's natural curiosity and exploratory drive entirely onto the drug cue.

- **The Opioid Link**: Natural social bonding relies on the brain's internal opioid networks. When social bonds fracture, the PANIC/GRIEF network fires, generating the agonizing emotional pain of loneliness. Panksepp demonstrates that addictive drugs (like opiates) chemically mimic these internal systems, serving as a desperate artificial attempt to soothe missing or fractured human connections.

- **The Control-Theoretic Interperatation:** In our control-theoretic terms, this subcortical hijacking manifests as extreme attentional capture ($\kappa \to 1$), a rigidly positive orientation toward the drug cue ($\Omega > 0$), and a catastrophic collapse of the participation ratio ($D_{\text{eff}} \to 1$). The organism's behavioral degrees of freedom shrink until the entire choice architecture is enslaved by a single, inescapable compulsive attractor sink.

5. Implications for Artificial General Intelligence and Machine Sentience

Using this theory, a homeostatic control system plus a self-model of its own control state is necessary; sufficiency is open. As a prediction of this framework, running standard transformer models does not generate subjective experience.

5.1 Why Pure Transformer Architectures Are Not Sentient

Current state-of-the-art language models lack foundational prerequisites for subjective experience:

1. **Absence of Homeostatic Layer:** LLMs do not maintain an internal, physical body state whose failure results in structural death or homeostatic collapse
2. **Abstract vs. Embodied Action Readiness:** Linguistic outputs expressing distress are semantic associations, not genuine action readiness vectors (**A**) constraining physical survival.
3. **The Missing Self-Model Description:** Without an architectural self-model compressing real-time somatic control vectors into a centralized description, there is no integrated coordinate space.

5.2 Architecting Machine Sentience

If elementary subjective experience is the informational geometry of self-correcting homeostatic control loops navigating survival, then carbon versus silicon is just a hardware implementation detail. To engineer a system capable of subjective feeling under this framework, an architecture must have essential variables H within a viability region $\mathcal{V}$, an active cost-benefit valuation ($\mathrm{SV} = R - E$), and a lossy coarse-graining self-model. $\mathcal{V}$ is the viability region, outside which the organism cannot recover and whose violation is irreversible for the system.

Traditional LLMs have no felt valence because their valuation and evaluation is done in the training and learning phase. After the training phase their weights are fixed. **Continuous learning** changes the situation. This new kind of model needs a reinforcement or reward system, which update the weights dynamically. These AIs must have an internal self-evaluation, and this self-evaluation can be the basis of subjective experience.

5.3 Ethical Implications

If the theory is right, plausibly, sentience and affective valence are mechanically coupled to vulnerability. Engineering genuine subjective well-being in a machine may require incorporating architectures capable of registering catastrophic harm, raising profound ethical questions regarding artificial suffering. Real subjective experience as humans have would mean real joy, but also real suffering.

6. Conclusion and Future Directions

6.1 Core Contributions of the Framework

The framework tries to build a bridge between philosophy of mind, psychology, biology, ethology and neuroscience, and hopefully brings us a step closer to solving the explanatory gap of the hard problem of consciousness (Levine, 2001). It suggests that the elementary subjective experience of feelings is based on the shape of the organism's control strategy as seen by its recursive self-model**.** Elementary feelings are, in this model, the lowest layer of consciousness itself: the self-model's pre-reflective awareness of the organism's own situation, given through the geometry of its control loops, as pressing and blocked or open and eased.

We describe affects as geometric shapes in the behavior space that constrain or enable action. Pain is a STOP constraint with attentional capture, hunger a pull, fear a push, relief a release of pressure, and grief a pull toward an unreachable target. These are recognizable from the inside, and they explain structurally why physical and social pain feel alike.

Following in the footsteps Antonio Damasio, Michael Graziano, Mark Solms and Nicholas Thompson we propose that elementary feelings are identical to a minimal form of elementary affective consciousness: the self-model's representation of the organism's own situation through the geometry of an active homeostatic control loop, as urgent or relieved, blocked or open, pulling or pushing. This elementary affective consciousness is pre-reflective and situated, and requires no narrative self, body concept or reflection.

If the identity holds, there is no further fact about what these feelings are like beyond the represented geometry, and the explanatory gap closes for this elementary domain, while the sensory qualities of the raw input data and the reflective, conceptual and social layers that build on it remain open. The irreducible core dimensions of qualia (what things feel like) reflect the orientation vectors of physical self-preservation in behavior space.

6.2 Future Directions for Research

- Building multi-agent neural network simulations governed by explicit homeostatic constraints to observe emergent participation ratio dynamics.

- Using statistical field theory and Renormalization Group (RG) dynamics (Wilson, 1975). Statistical field theory is a theoretical framework in physics that describes systems with a large or infinite number of interacting degrees of freedom by replacing discrete microscopic variables with continuous fields. RG flows are the trajectories or paths that describe how the effective description of a physical system - such as its parameters or interactions - changes as you zoom in or out across different length or energy scales. RG flows often show a dynamic where fixed points behave as saddle points. It has been applied to deep learning (Mehta & Schwab, 2014), and it is possible that neural data shows RG scaling with non-trivial exponents, as Meshulam et al. suggested (Meshulam, 2019)

6.3 Final Remarks

Staring directly at the mechanics of the own existence can be frightening, but we hope it wasn't. Elementary subjective consciousness is understood here simply as the self-model's lossy description of an organism steering its own survival through space and time. Reframing qualia as the informational geometry of self-correcting control loops provides a clear, testable roadmap for future cognitive science.

You may ask but why should lossy compression "feel like anything" and have a certain quality? We can use a thermodynamic analogy to show that a macroscopic quantity can emerge by coarse-graining when micro-details are vanishing and high-level structures appear. If heat, cold and temperature in general are identified with average molecular kinetic energy and viscosity (resistance to flow, the opposite of fluidity) is mean molecular stickiness, then temperature and viscosity are a lossy compression of molecular activity. Therefore, lossy compression is not only useful, but it could also be necessary to describe emergent properties on a higher level. We accept without hesitation that macroscopic properties like temperature and viscosity are real, physical properties of matter *because* of coarse graining. We view the world from a coarse-grained point of view, not from a molecular level, and we know these macroscopic properties can be defined as a lossy compression of microscopic molecular activity

We propose that subjective quantities like pain are similar: macroscopic properties of a large neural system that emerge as high-level macroscopic properties by coarse-graining its microscopic activity, in which the details of individual neural activity vanish. A subjective experience like pain is similar to a temperature. Pain intensity is routinely measured in hospitals by asking a patient to rate it on a scale from 0 to 10. A rich subjective state is reduced to a single number. For temperature we do the same. We first knew it as a felt quality, hot and cold, then quantified it with a thermometer, and found out we can identify it with the average kinetic energy of molecules. In that average all molecular details vanish, yet nobody thinks temperature is unreal or something additional to molecular motion.

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